MA411 Notes 6

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16 Martingale representation, stopping time

Let $(\Omega, \mathcal{F}, \mathbb{P})$ with $(\mathcal{F}_n)_{n\geq 0}$ be a filtered probability space. Suppose $\sum_{i=1}^n Y_i$ is a *reference* martingale by summation (i.e. $\mathbb{E}(Y_i|\mathcal{F}_{i-1}) = 0$ for all i, and the random variables are in L_1): this means $\mathcal{F}_n = \sigma(Y_1, \dots, Y_n)$ holds for each $n \in \mathbb{N}$.

Suppose now we are given another martingale $(X_n)_{n\geq 0}$ adapted to the same filtration. Can we describe this new martingale in terms of our reference martingale?

Definition 1 (Martingale transform)**.** *Given $(\Omega, \mathcal{F}, \mathbb{P})$ with $(\mathcal{F}_n)_{n\geq 0}$ a filtered probability space, let $\sum_{i=1}^n Y_i$ be a martingale by summation adapted to the filtration. Let $(C_n)_{n\geq 0}$ be any process adapted to the filtration. Then the* martingale transform *is $\sum_{i=1}^n C_{i-1}Y_i$.*

It's (somewhat) standard to write $C_1, C_2, \dots$ and define C_i to be $\mathcal{F}_{i-1}$-measurable ('previsible process'). If you do this, then the indices will all be i in the martingale transform, but you have to remember that the measurability and the indices do not match. I prefer to have the index always indicate the measurability; this has the advantage that I don't need to define a new word for $(C_0, C_1, \dots)$ too.

It's also quite standard to define the martingale transform by starting with a martingale $(Z_n)_{n\geq 0}$ and define $Y_n = Z_n - Z_{n-1}$ as the *martingale difference sequence*. This is of course equivalent to the way we defined it.

We saw in class that if $\|C_n\| < \infty$ for each $n \geq 0$, then the martingale transform is a martingale. Here is a generalisation.

Exercise 1. *Suppose each Y_i is in $L_p(\Omega, \mathcal{F}, \mathbb{P})$, and each C_i is in $L_q(\Omega, \mathcal{F}, \mathbb{P})$, where $1 \leq p, q \leq \infty$ and $\frac{1}{p} + \frac{1}{q} \leq 1$. Suppose $\mathbb{E}(Y_n|\mathcal{F}_{n-1})] = 0$ for all $n \geq 0$. Prove that $\sum_{i=1}^n C_{i-1}Y_i$ is a martingale adapted to $(\mathcal{F}_n)$.*

Solution to Exercise 1:

The proof that the sum is adapted to the filtration and of the martingale property are as we saw in class. That the $\sum_{i=1}^n C_{i-1}Y_i$ is $\mathcal{F}_n$-measurable for each $n \geq 0$ follows since each term in the sum is by definition measurable with respect to some $\mathcal{F}_j$ with $j \leq n$, so also with respect to $\mathcal{F}_n$ by definition of 'filtration': and finite sums and products of $\mathcal{F}_n$-measurable functions are $\mathcal{F}_n$-measurable.

For the martingale property, we have

$$\begin{aligned}\mathbb{E}\Big(\sum_{i=1}^{n} C_{i-1}Y_i\big|\mathcal{F}_{n-1}\Big) &= \mathbb{E}(C_{n-1}Y_n|\mathcal{F}_n) + \mathbb{E}\Big(\sum_{i=1}^{n-1} C_{i-1}Y_i\big|\mathcal{F}_{n-1}\Big)\\ &= C_{n-1}\mathbb{E}(Y_n|\mathcal{F}_{n-1}) + \mathbb{E}\Big(\sum_{i=1}^{n-1} C_{i-1}Y_i\big|\mathcal{F}_{n-1}\Big)\\ &= \mathbb{E}\Big(\sum_{i=1}^{n-1} C_{i-1}Y_i\big|\mathcal{F}_{n-1}\Big)\end{aligned}$$

where the first equality is linearity of the conditional expectation, the second is 'taking out what is known', and the third is by assumption on the conditional expectation.

Finally, we need the L_1 property: this is where we have to do something new. For $n = 0$ we have the empty sum; the constant zero function is in L_1 trivially. We prove the rest by induction. Let $p' \leq p$ be such that $\frac{1}{p'} + \frac{1}{q} = 1$; since $\|Y_n\|_p < \infty$ we also have $\|Y_n\|_{p'} < \infty$.

We have

$$\Big\|\sum_{i=1}^{n} C_{i-1}Y_i\Big\|_1 \leq \Big\|\sum_{i=1}^{n-1} C_{i-1}Y_i\Big\|_1 + \|C_{i-1}Y_i\|_1 \leq \Big\|\sum_{i=1}^{n-1} C_{i-1}Y_i\Big\|_1 + \|C_{n-1}\|_q\|Y_n\|_{p'} < \infty\,,$$

where the first inequality is the triangle inequality, the second is Hölder's inequality, and the three norm terms on the RHS are finite respectively by induction, and by assumption of L_q- and $L_{p'}$-containment of C_{n-1} and Y_n.

We saw in Notes 2 that, since X_n is $\sigma(Y_1, \ldots, Y_n)$-measurable, we can write $X_n = f_n(Y_1, \ldots, Y_n)$. In general, this is the closest we can come to a 'converse' to Exercise 1; we cannot in general expect to write $(X_n)_{n\geq 0}$ as a martingale transform. We don't even really need to bring in the whole martingale formalism to see this; things go wrong already at the first step. If Y_1 takes values in $\{-1, 0, 1\}$ with equal probability (which means its expectation is zero), then for instance we could have X_1 taking value -2 if $Y_1 = -1$ and 1 otherwise. Then X_1 is a function of Y_1, so it's $\sigma(Y_1)$-measurable, and $\mathbb{E}(X_1) = -2\cdot\frac{1}{3}+1\cdot\frac{2}{3} = 0$, but we obviously can't write $X_1 = cY_1$ for any constant c.

However, there is a special case when we do get a converse, which is the simple random walk and some generalisations.

Theorem 2. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space; let $(Y_n)_{n\in\mathbb{N}}$ be independent uniform random variables taking values in $\{-1, 1\}$, and let $(\mathcal{F}_n)_{n\geq 0}$ be the natural filtration of the process. Then for every $(\mathcal{F}_n)_{n\geq 0}$-adapted martingale $(X_n)_{n\geq 0}$, there exists a unique $(\mathcal{F}_n)_{n\geq 0}$-adapted process $(C_n)_{n\geq 0}$ such that $X_n = X_0 + \sum_{i=1}^{n} C_{i-1}Y_i$ for all $n \in \mathbb{N}$.*

Proof. The setup invites a proof by induction: suppose $C_0, \ldots, C_{n-2}$ have been defined and are unique such that $X_m = X_0 + \sum_{i=1}^{m} C_{i-1}Y_i$ holds for $m = 0, \ldots, n-1$. We now aim to define a suitable C_{n-1}.

Observe that $\mathcal{F}_n$ is a finite σ-algebra for each n, so conditional probabilities are uniquely defined and given simply by conditioning on the minimal non-empty events. Furthermore it makes sense to talk about the *conditioned random variable* $X_n|F_{n-1}$, where F_{n-1} is a minimal

non-empty set in $\mathcal{F}_{n-1}$. By this we mean the random variable $X_n|_{F_{n-1}}$, with F_{n-1} given the *conditioned measure* $\mathbb{P}_{F_{n-1}}(A) = \mathbb{P}(A)/\mathbb{P}(F_{n-1})$ for all $A \in \mathcal{F}$ which are contained in F_{n-1}. (This concept of course makes sense for any set F_{n-1} with non-zero measure, and it is a standard definition, but it's only useful for dealing with martingales in rather special cases, such as this one.)

Because Y_n has only two possible values, for each minimal non-empty set $F_{n-1} \in \mathcal{F}_{n-1}$, the conditioned random variable $X_n|F_{n-1}$ can be written as $X_n|F_{n-1} = f_{n-1,F_{n-1}}(Y_n)$ for some function $f_{n-1,F_{n-1}}$, and therefore takes only two values. By the martingale property, $\mathbb{E}(X_n|F_{n-1}) = X_{n-1}(\omega)$ for any choice of $\omega \in F_{n-1}$, and so the two values are $X_{n-1}(\omega) + c_{n-1,F_{n-1}}$ (if $Y_n = 1$) and $X_{n-1}(\omega) - c_{n-1,F_{n-1}}$ (if $Y_n = -1$), for some constant $c_{n-1,F_{n-1}}$: here we use the subscript to highlight that the constant can in general depend on F_{n-1} and on $n-1$. We define a random variable C_{n-1} by $C_{n-1}(\omega) = c_{n-1,F_{n-1}}$ where F_{n-1} is the minimal non-empty set of $\mathcal{F}_{n-1}$ containing ω. It's clear that this was the only possible definition: if we replaced C_{n-1} by some other C'_{n-1}, then it would have to differ from C_{n-1} on some minimal non-empty $F_{n-1} \in \mathcal{F}_{n-1}$, and since $Y_n \neq 0$ on this set, the result is a different value for $\sum_{i=1}^{n-1} C_{i-1}Y_i + C'_{n-1}Y_n$, so that this sum cannot be almost surely equal to X_n.

By definition C_{n-1} is $\mathcal{F}_{n-1}$-measurable, and by construction we have $X_n = X_0 + \sum_{i=1}^{n} C_{i-1}Y_i$, as required. This completes the proof by induction. $\square$

Exercise 2. *Generalise Theorem 2: prove that we get the same conclusion if $Y_n|F_{n-1}$ is a random variable which takes at most two values and has expectation zero for every minimal non-empty $F_{n-1} \in \mathcal{F}_{n-1} = \sigma(Y_1, \ldots, Y_{n-1})$. If (and only if) $Y_n|F_{n-1}$ takes exactly two distinct values for all minimal non-empty $F_{n-1} \in \mathcal{F}_{n-1}$, for all $n \in \mathbb{N}$, then $(C_n)_{n\geq 0}$ is unique.*

Solution to Exercise 2:

The proof of Theorem 2 more or less generalises; again, we perform induction on n and suppose $C_0, \ldots, C_{n-2}$ are defined with the required property.

Observe that if $Y_n|F_{n-1}$ takes only one value for some F_{n-1} a minimal non-empty set in $\mathcal{F}_{n-1}$, then that value is necessarily zero and in this case F_{n-1} is also a minimal non-empty set in $\mathcal{F}_n$, so X_n is constant on F_{n-1} and by the martingale property we have $X_n = X_{n-1}$ on $\mathcal{F}_{n-1}$. It follows that whatever choice we make for C_{n-1} on the set F_{n-1} we obtain (on this set!) the required transform property; for completeness, let's say that in this case we will choose to take $C_{n-1} = 0$ on F_{n-1} (but note that we did have a choice!).

Now suppose F_{n-1} a minimal non-empty set in $\mathcal{F}_{n-1}$ is such that $Y_n|F_{n-1}$ takes two distinct values. Suppose these are $z_{n-1,F_{n-1}}$ and z', with non-zero probabilities respectively $p_{n-1,F_{n-1}}$ and $1 - p_{n-1,F_{n-1}}$. Necessarily we have $z'(1 - p_{n-1,F_{n-1}}) + z_{n-1,F_{n-1}}p_{n-1,F_{n-1}} = 0$, because $\mathbb{E}(Y_i|F_{n-1}) = 0$.

As before, $X_n|F_{n-1}$ is a function of $Y_n|F_{n-1}$; it takes (at most) two values, with the same probabilities, so these values are $x + y_{n-1,F_{n-1}}$ and $x + y'$, where we have $x = \mathbb{E}(X_n|F_{n-1})$ and $y_{n-1,F_{n-1}}p_{n-1,F_{n-1}} + y'(1 - p_{n-1,F_{n-1}}) = 0$. We let $C_{n-1}(\omega) = y_{n-1,F_{n-1}}/z_{n-1,F_{n-1}}$, and observe that this gives $y' = C_{n-1}(\omega)z'$. As before, C_{n-1} is $\mathcal{F}_{n-1}$-measurable by construction, and by construction we obtain $X_n = X_0 + \sum_{i=1}^{n} C_{i-1}Y_i$, and we had no choice in defining C_{n-1} on F_{n-1}.

If you have seen the Riemann-Stieltjes integral, you might recognise that a martingale transform looks a bit like a Riemann-Stieltjes integral with respect to the martingale $\sum_{i=1}^{n} Y_i$.

If you have seen the Itô integral, you should see that what is going on is something like a discretisation of the Itô integral: we're multiplying the process value C_{i-1} by the martingale difference Y_i, instead of (some kind of limit of!) the difference between two points on a Brownian motion (or whatever you're Itô-integrating with respect to).

To make this connection more precise, suppose you want to integrate a simple process with respect to Brownian motion, and the simple process happens to have quite long times between its jumps (recall a simple process is constant between fixed time jumps, though the values it takes at each jump are random and history-dependent). The difference in the Brownian motion values from one jump point to the next will be a Gaussian random variable: and the simple random over a long time will give a distribution which also looks close to Gaussian. In fact, the way to make this connection more precise is not to insist on long times between jumps, but to fix a simple process and then to consider a *rescaled* simple random walk which jumps at every $1/N$ units of time, and up or down by $1/\sqrt{N}$ not by 1; as N grows, for a fixed simple process, the change in the rescaled simple random walk approaches the 'correct' Gaussian change of the Brownian motion (this is the Central Limit Theorem), and intuitively the martingale transform approaches the Itô integral; we can view the latter as (some kind of!) 'continuous limit' of the former. Intuitively, the same should be true if we drop the 'simple' requirement. However, this is *not* how the Itô integral is normally defined: unless you want to do rather a lot of technical work, it's best to treat this as an intuitive view and not a rigorous statement.

Nevertheless, this gives a good intuition as to why the continuous-time martingale representation 'should' be true—which is something you should distrust after seeing that the discrete version fails as soon as we have three-valued random variables!—which is that the Brownian motion (and other related processes with respect to which martingale representation holds) happens to be some kind of 'limit' of a martingale process for which the discrete martingale representation does hold, namely the simple random walk.

Finally, we mention one concept which 'should' appear in a Probability and Measure course, even though we're not going to use it.

Definition 3 (stopping time)**.** *Given a filtered probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *with filtration* $(\mathcal{F}_n)_{n\geq 0}$*, a discrete* stopping time *is a random variable* $\tau : \Omega \to \mathbb{Z}_{\geq 0} \cup \{\infty\}$ *such that* $\{\omega : \tau(\omega) = n\}$ *is in* $\mathcal{F}_n$ *for each* $n \geq 0$*.*

Intuitively, what this definition means is that you have a process, and you want to 'stop' if some event occurs. This will only be possible if the event doesn't require you to predict the future. So you cannot stop at 'one minute before the market collapses'; you don't know the market collapse will happen until it does. You can, on the other hand, stop at 'the first time the price hits £200'.

The constant function taking any given value in $\mathbb{Z}_{\geq}0$ is a stopping time. For any martingale $(X_n)_{n\geq 0}$ adapted to the filtration, and any $c \in \mathbb{R}$, the function $\tau(\omega)$ defined to equal the first n such that $X_n(\omega) \geq a$ is a stopping time (called a *hitting time*). This remains true if 'first' is replaced by 'second' and so on, but not 'last'.

Exercise 3. *Explain why the above examples are (for 'last', is not in general a) stopping times.*

Solution to Exercise 3:

The constant c function (where $c \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ is easily seen to be a stopping time: indeed the event $\{\tau = n\}$ is equal to either Ω or $\emptyset$, depending on whether $n = c$ or not. Both these events are in $\mathcal{F}_0$ and so also in $\mathcal{F}_n$.

For a hitting time, given $(X_n)_{n\geq 0}$ and a real number $a > X_0$ (the case $a < X_0$ is similar!), consider the hitting time 'the first time $X_n \geq a$'. Observe that for a given n, the event $\tau = n$ is the same as the event

$$X_0, \ldots, X_{n-1} < a \quad \text{and} \quad X_n \geq a\,.$$

Since $\{X_i < a\}$ is $\mathcal{F}_i$-measurable for each $0 \leq i \leq n-1$, it is also $\mathcal{F}_n$-measurable because $\mathcal{F}_i \subseteq \mathcal{F}_n$, and by definition $\{X_n \geq a\}$ is $\mathcal{F}_n$-measurable. Now $\{\tau = n\}$ is the intersection of these events, which is $\mathcal{F}_n$-measurable since $\mathcal{F}_n$ is a σ-algebra. This verifies this τ is a stopping time.

A similar argument deals with 'second time', or 'third time', or even '$f(n)$th time', where $f(n)$ is allowed to grow with n; you just list off all the possible ways that $\tau = n$ could occur, in terms of the X_i being either $< a$ or $\geq a$, and observe there will be finitely many possibilities each of which is $\mathcal{F}_n$-measurable.

However 'last time' doesn't work. It's very tempting to say that a simple random walk with $a = 1$ provides an example, because whatever n you declare to be the 'last time' the simple random walk exceeds 1, it's still possible for the simple random walk to return to 1 at some future time. But in fact, for this example there *is* a stopping time: $\tau = \infty$ is almost surely the 'correct' stopping time; with probability 1 the simple random walk will return to 1 infinitely often. Maybe the simplest example is the halting walk. Consider the process which for the first 100 steps follows the simple random walk, and thereafter does not change; in other words $X_n = \sum_{i=1}^{\min(n,100)} Y_i$ where each Y_i is independently and uniformly chosen from $\{-1, 1\}$. It's easy to check this process is a martingale. However 'the last time the process is at least 1' is not a hitting time. With probability $\frac{1}{2}$ the process will be at 1 at time 1; let A be the subset of Ω such that $X_1 = 1$. Since $\mathcal{F}_1 = \{\emptyset, A, \Omega \setminus A, \Omega\}$, if τ is to be any stopping time, then $\tau(\omega) = 1$ for either all, or no, $\omega \in A$. But it's easy to see (by direct computation, say) that for some $\omega \in A$ the process will be above 1 at a future time (for example, for half of A this will be the case at time 2) while for others the process remains below 1 for all $t \geq 2$ (for example the subset of A where $Y_i = -1$ for all $2 \leq i \leq 100$); both those sets have positive measure, but for the first case 'the last time' is at least 2, while for the second 'the last time' is 1.

If τ is a stopping time and $(X_n)_{n\geq 0}$ is a martingale (both adapted to the same filtration) then X_τ is a well-defined random variable. Intuitively, it seems reasonable that $\mathbb{E}(X_\tau) = X_0$.

However, this is famously false. Consider the martingale betting strategy: double-or-quits until you are ahead. That is, let $Y_1, \ldots$ be independent random variables taking values uniformly in $\{-1, 1\}$, and let $X_n = \sum_{i=1}^{n} 2^{i-1} Y_i$. Let $\tau(\omega)$ be the first n such that $X_n \geq 1$.

In this example, it's easy to check that $\tau(\omega)$ will be the first n such that $Y_n = 1$, and X_τ is almost surely equal to 1; in particular its expectation is not zero.

The *optional stopping theorem* gives some conditions for X_τ to exist and for $\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$ to hold. In particular, this is true whenever there exists a constant c such that $\tau \leq c$ holds almost surely.

The 'X_τ' exists' part is the observation that if $\tau < \infty$ does not hold almost surely, then it's unclear how X_τ should be defined. We can deal with $\tau = \infty$ holding on a set of measure zero 'arbitrarily', for example by defining $X_\tau = 0$ if $\tau = \infty$; we're only changing a set of measure zero. But if $\tau = \infty$ on a set of positive measure, we would likely need to use some kind of limit to define X_τ, and whether this limit preserves 'good properties' isn't always clear.

Exercise 4. *Prove that if there is c such that $\tau \leq c$ holds almost surely and τ is a stopping time, then X_τ exists and has expectation $\mathbb{E}(X_0)$. Hint: consider the* stopped process *defined by $(Y_n)_{n\geq 0}$ by $Y_n(\omega) = X_n(\omega)$ whenever $n \leq \tau(\omega)$ and $Y_n(\omega) = X_{\tau(\omega)}(\omega)$ whenever $n > \tau(\omega)$. Show that this is a martingale, and use this to prove the required statement.*

Solution to Exercise 4:

The existence is clear: for ω such that $\tau(\omega) < \infty$, we have $X_\tau = X_{\tau(\omega)}(\omega)$. Since this holds almost surely, when $\tau = \infty$ we can set $X_\tau = 0$.

We claim the stopped process is $(\mathcal{F}_n)$-adapted. Indeed, we can write

$$Y_n = X_n \mathbf{1}_{\tau \geq n} + \sum_{t=0}^{n-1} X_t \mathbf{1}_{\tau = t}\,,$$

and by definition of a stopping time, all the terms in this sum are $\mathcal{F}_n$-measurable. From the same summation formula, monotonicity of the integral, and the triangle inequality, we get

$$\mathbb{E}(|Y_n|) \leq \mathbb{E}(|X_0|) + \cdots + \mathbb{E}(|X_n|) < \infty\,.$$

Finally, we observe the stopped process is a martingale. Indeed, again from the summation formula we have

$$\begin{aligned}
\mathbb{E}(Y_{n+1}|\mathcal{F}_n) &= \mathbb{E}(X_{n+1}\mathbf{1}_{\tau\geq n+1}|\mathcal{F}_n) + \sum_{t=0}^{n} \mathbb{E}(X_t \mathbf{1}_{\tau=t}|\mathcal{F}_n) \\
&= \mathbb{E}(X_{n+1}|\mathcal{F}_n)\mathbf{1}_{\tau\geq n+1} + \sum_{t=0}^{n} X_t \mathbf{1}_{\tau=t} \\
&= X_n \mathbf{1}_{\tau\geq n+1} + X_n \mathbf{1}_{\tau=n} + \sum_{t=0}^{n-1} X_t \mathbf{1}_{\tau=t} \\
&= X_n \mathbf{1}_{\tau\geq n} + \sum_{t=0}^{n-1} X_t \mathbf{1}_{\tau=t} = Y_n\,,
\end{aligned}$$

as required. Here the second equality is 'taking out what is known': observe $\mathbf{1}_{\tau\geq n+1} = 1 - \mathbf{1}_{\tau\leq n}$ is $\mathcal{F}_n$-measurable, and the third equality is the martingale property of X_{n+1}.

Since $\tau \leq c$ almost surely, $Y_c = X_\tau$ almost surely. So $\mathbb{E}(X_\tau) = \mathbb{E}(Y_c) = \mathbb{E}(Y_0) = \mathbb{E}(X_0)$, as required.

Two other standard conditions for X_τ to exist with $\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$ are

- $\mathbb{E}(\tau) < \infty$ and there is c such that for all t, within $\{\tau \geq t\}$ we have $\left|\mathbb{E}(X_{t+1} - X_t|\mathcal{F}_t)\right| \leq c$. (The latter can be slightly more formally phrased as: if ω is such that $\tau(\omega) \geq t$, then $\left|\mathbb{E}(X_{t+1} - X_t|\mathcal{F}_t)(\omega)\right| \leq c$.)

- There is c such that $\left|X_{\min(t,\tau)}\right| \leq c$ holds almost surely for all t.

In the former case, by Markov's inequality almost surely $\tau < \infty$ and it is not too hard to prove the corresponding expectation statement. In the latter case, we can have $\mathbb{P}(\tau = \infty) > 0$; we need to define X_τ to be the pointwise limit of X_t where it exists, and we need to prove that this 'where it exists' is almost surely; this needs some more work.

These three conditions together are usually called the Optional Stopping Theorem. We won't need this result (we could argue we use the case we proved below, but the case for a fixed constant stopping time is rather trivial!); but if you say you've taken a course in Probability and Measure, it will likely be assumed you saw it. Mainly, what you should notice is that the positive conditions are quite restrictive; a lot of 'natural' stopping times, such as most hitting times, don't satisfy these conditions and also (there is no if-and-only-if here!) don't satisfy the expectation property. It's obvious that a hitting time (say, of the simple random walk starting from 0, hitting 1) won't satisfy the expectation condition (in this example, $X_\tau = 1$ almost surely..!), but it's maybe not obvious why none of the three conditions apply. Well, the stopping time is not almost surely bounded—random-walking only downwards for any given number of steps has positive probability—and the martingale is obviously not almost surely bounded for the same reason. You might well think the expected stopping time is finite; after all, it is quite likely that we hit 1 quickly. But if you try to do the computation properly, you'll discover this one is false; the expectation of the hitting time is ∞.

17 The binomial model

In this section we give a simple discrete financial model. We'll consider a market in which there is only one stock, whose value changes at discrete time intervals; as the player in the market, your actions are modelled as having no effect on the price change. Furthermore, we'll restrict the value changes to be two possibilities: the value either goes up or down by fixed fractions: we'll return to why this might be a sensible thing to do later. In order to make this 'realistic', you should think of one unit of time as being a very small fraction of a second. We'll assume you can buy and sell any amount of the stock you like, at its current price, for no fee ('frictionless trading'); in particular, that means you are allowed to own fractional quantities of the stock, and you are permitted to own negative amounts of the stock. The latter is realistic; it is called 'a short position' and it simply means that you owe the market stock; 'in the future' you will have to buy the stock until you pay off your debt. The former is not very realistic—mainly because if you genuinely could buy or sell arbitrary amounts, you certainly would change market prices: but in practice the strategy that is coming up won't require you to buy or sell very large amounts.

Finally, we'll assume that whatever money you currently hold will be invested at a fixed interest rate. The 'fixed' part is unrealistic; but it makes things much simpler (and it is not too unrealistic, at least if we're thinking about short-ish intervals of real time). Traditionally 'invested money' is called *bonds*; to conform to the literature (and because this notation is easier to work with if you make a more realistic model!) we'll think about a 'bond price' which increases by a fixed fraction every time step. At time 0, the stock price is $S_0 > 0$ and the bond price is $B_0 > 0$. We are given fixed numbers $-1 < a < r < b < \infty$, and at each

time step $t \geq 1$, these numbers evolve as follows:

$$B_t = B_{t-1}(1+r) \quad \text{and} \quad S_t = S_{t-1} \cdot \begin{cases} (1+a) & \text{if } Y_t = -1 \\ (1+b) & \text{if } Y_t = 1 \end{cases} .$$

So r is the fixed interest rate; and $1+a$ and $1+b$ are the two possible fixed fractions by which the stock price changes. We'll think of $1+a$ as 'a jump down', and $1+b$ as 'a jump up', even if formally these two numbers might not be either side of 1. The justification is that the effective value of money is changing by $1+r$ every time step—that's what you would get if you simply parked your wealth in the risk free bond—and relative to that the two numbers really are respectively worse and better.

The big assumption in this model is the choice of a and b. At the moment, it's not clear where these numbers come from; they're simply given to us. We'll see later what they represent. The interest rate r is fairly clear: at any given time, you can ask your bank (or the bond market!) what the current interest rate is.

We should be clear that right now Y_n is simply a function from some set Ω to $\{-1, 1\}$ which we use to keep track of how the stock price changes over time; we haven't even defined a probability measure, so no randomness has come in to the picture just yet. However it's convenient, for now purely notationally, to let $\mathcal{F}_n = \sigma(Y_1, \ldots, Y_n)$ so that we have a filtration on the measurable space $(\Omega, \mathcal{P}(\Omega))$.

A *European option* is a financial contract, which can be purchased at a cost. It comes with a *maturity time* N and a *strike price* K. If you purchase the option you can, if you choose, buy a unit of the stock at the fixed price K, at time N.

It's best to think of the option as something offered to you by another player in the market. Your actions don't influence theirs, so they are not trying to manipulate you into any specific action. They simply hope that you will not know what a fair cost is and hence pay a cost which is higher than you 'should' pay, so that they get free money right now—after that, they have no interest in what you do.

If you purchase the option, at time N it is clear what you will do: if $S_N \geq K$, you will call in the option and buy a unit of stock at price K, and if $S_N < K$, you will forfeit the option (the other player gets to keep the purchase cost in either case, of course).

However, at time 0 neither you nor the other player can see the future. The other player, who wants 'free money', is considered to be risk averse: they want a guarantee that if you overpay, they will definitely profit. They are worried that for some $\omega \in \Omega$ (some possible future), we might have S_N significantly larger than K, and if they do nothing beyond sell you the option, they stand to lose money. They are, again, not interested in the probability of ω happening, and in our model we don't know that probability; any possibility of loss is unacceptable.

A *hedging strategy* for the option, with initial value x, is a process $(A_n, V_n)_{n \geq 0}$ which is adapted to $(\mathcal{F}_n)_{n \geq 0}$, and which satisfies:

- $x = A_0 S_0 + V_0 B_0$,
- for each $n \geq 0$, we have $A_n S_{n+1} + V_n B_{n+1} = A_{n+1} S_{n+1} + V_{n+1} B_{n+1}$, and

- $A_{N-1}S_N + V_{N-1}B_N = \max(S_N - K, 0)$.

The idea here is that immediately after time n (having seen the stock price S_n), the player makes a decision to buy or sell stock, so that their holding of stock immediately after time n is A_n. The rest of their wealth goes into bonds, of which they hold V_n units. So the first bullet point represents the player at time 0, with wealth x, apportioning that wealth between stocks and bonds. At time 1, their wealth has changed as the stock and bond prices update: B_n is deterministic, but S_n updates unpredictably. The equality of the second bullet point represents the player, immediately after time $n+1$, making a trade: they buy (or sell if negative!) $A_{n+1} - A_n$ units of stock, at the current stock price S_{n+1}, and this must be matched by a change in their holdings of the bond: the player's wealth is not changed by the trade. Finally, the last bullet point represents the player's wealth at time N, once the stock prices at time N are known: and the requirement is that this wealth is identical to the player's wealth if they had instead bought for a cost of x the option at time 0.

Again, we should stress that there is nothing random about this; we use the probabilistic formalisms, but just as a convenient way of keeping track of information. In particular, the requirement that $(A_n, V_n)_{n\geq 0}$ is $(\mathcal{F}_n)_{n\geq 0}$-adapted simply says that the player's decisions to trade immediately after time n can depend only on the stock price changes they have seen up to and including time n, and not on any future changes. We should also point out that (unlike the usual financial literature) our subscripts always indicate the measurability: if the subscript is at most n, then the random variable is $\mathcal{F}_n$-measurable. (If you read financial stuff, you will likely see A and V being given indices one larger than we do, but then being called 'pre-visible' to indicate that the index is one off from the measurability.)

The 'point' of this definition is the following: provided a hedging strategy exists with initial value x, the only possible fair price for the option is x. To see this, suppose you are offered the option at some cost $x' > x$. If you purchase, the player offering you the option can take your x', set aside $x' - x$ into bonds, and then with the remaining x buy stocks and bonds according to the hedging strategy. At time N, if the stock price is below K their wealth will consist of just the $(1+r)^N(x'-x)$ set aside in bonds, but you will not want to exercise your option, so the other player has made free money. If on the other hand the stock price is $S_N \geq K$, then you will want to exercise your option. But the other player's wealth is $(1+r)^N(x'-x) + S_N - K$. With the K you pay them to exercise the option, they have exactly the S_N needed to buy you one unit of the stock and still have $(1+r)^N(x'-x)$ over. Either way, they made free money, because you overpaid. (You would have been better off to follow the hedging strategy yourself.) Similarly, if the option is offered to you at some cost $x' < x$, then you should borrow x, by taking a A_0 short position on the stock (i.e. own $-A_0$) and V_0 short on bonds (i.e. you own $-V_0$ of bonds), buy the option and set aside the extra $x - x'$ in bonds. You then follow the hedging strategy in your short position (i.e. you just take the negative of the hedging strategy) to time N, and much as before you either find that $S_N < K$, but the hedging strategy has got you out of debt and you can happily keep the $(x-x')(1+r)^N$ set aside in bonds, or $S_N \geq K$, you still are in debt, but by exercising your option you can clear the debt and still keep the $(1+r)^N(x-x')$ set aside.

It's instructive to do this calculation by hand for $N = 1$.

Exercise 5. *Suppose $N = 1$. Given a, b, r, S_0 and B_0, determine a hedging strategy, and hence a fair price, for the option explicitly. Explain why the hedging strategy and fair price are unique.*

Solution to Exercise 5:

We need to choose A_0 and V_0, which are simply numbers, to satisfy $x = A_0S_0 + V_0B_0$ and $A_0S_1 + V_0B_1 = \max(S_1 - K, 0)$.

If $K > (1+b)S_0$, then the option will never be exercised and its fair price is 0; the hedging strategy is to buy nothing.

If $K < (1+a)S_0$, then the option is always exercised, and its fair price is $S_0 - K(1+r)^{-1}$; the hedging strategy is to buy one unit of stock ($A_0 = 1$) and take a short position $K(1+r)^{-1}B_0^{-1}$ ($V_0 = -L(1+r)^{-1}B_0^{-1}$) in bonds; at time 1 we then have the one unit of stock we need to give when the option is exercised, and we will receive K with which to pay our debt in bonds that is at this time exactly K.

So now suppose $(1+a)S_0 \le K \le (1+b)$. Then $A_0S_1 + V_0B_1 = \max(S_1 - K, 0)$ is really two equations:

$$A_0(1+a)S_0 + V_0(1+r)B_0 = 0 \quad \text{and} \quad A_0(1+b)S_0 + V_0(1+r)B_0 = (1+b)S_0 - K\,.$$

This is a pair of linear equations, and the relevant matrix $\begin{pmatrix} (1+a)S_0 & (1+r)B_0 \\ (1+b)S_0 & (1+r)B_0 \end{pmatrix}$ is non-singular (since $a \neq b$, the columns are linearly independent): so there is a unique solution for (A_0, V_0). Having found it, we plug it in to $x = A_0S_0 + V_0B_0$ to find the fair price x, which (since the hedging strategy is unique) is obviously unique. We can write out a formula for the fair price, but doing so isn't particularly instructive: the point is to note that this is a simple piece of linear algebra.

It's not too hard to see that we can iterate Exercise 5 to deal with $N > 1$. A good way to view what we did in Exercise 5 is that we had two possible values for the option at time N, and we computed its value at time $N-1$ from that.

In general, we have the *binomial tree* model: we draw a 'tree' starting with a node at time 0, with two 'child' nodes at time 1 according to the two possible changes in stock price, each of which have 2 children at time 2, and so on to time N where we have 2^N 'leaves', in each of which our option has a specific monetary value: then we determine as in Exercise 5 what hedging strategy and hence value we have for each of the time $N-1$ nodes.

We can then work backwards: given a time $N-2$ node, and its two time $N-1$ children, now that we have a value for the option at each of the child nodes, we can as in Exercise 5 compute the value of the option at the time $N-2$ node, and so on for all time $N-2$ nodes. Now we continue, working backwards, with the time $N-3$ nodes, and so on till we get to the time 0 node. This gives us a complete hedging strategy and a fair price at time 0; as in Exercise 5 the hedging strategy is unique.

However, in practice this approach is computationally difficult; if N is at all large, it's infeasible to perform the calculation as just described, because we need to do 2^N calculations, at least if we use the tree model as described.

Exercise 6. *This particular model is* recombining*, meaning that in fact the value of the option and hedging strategy at a given time t node depends only on S_t and not on the specific sequences of moves up and down to reach the node. Explain why this is the case. Hint: use induction, working backwards in time.*

Explain why this reduces the number of calculations required to obtain a complete hedging strategy and option value to something on the order of N^2.

Solution to Exercise 6:

Following the hint, we use induction. For the time N nodes, the claim is true by definition: there is no hedging strategy at this point and the value of the option is $\max(0, S_N - K)$ does not depend on the sequence of moves reaching S_N.

Suppose now that for some $0 \leq t < N$ we have established the claim for the time $t+1$ nodes and thereafter. Suppose some two time t nodes P and Q agree on the value of S_t.

The children of P, and the children of Q, are all time $t+1$ nodes. So the move-up-child of P and of Q both have stock price $(1+b)S_t$, and, because the binomial trees and hedging strategies are unique, they must agree on the value v_u of the option and the hedging strategy (i.e. the wealth required to hedge the option at that particular node). Similarly, there is agreement between the move-down-child of P and of Q on the value v_d and the hedging strategy.

To find the value of the option, and the hedging strategy, at node P, we solve for A_t and V_t

$$\begin{pmatrix} (1+a)S_t & (1+r)B_t \\ (1+b)S_t & (1+r)B_t \end{pmatrix} \begin{pmatrix} A_t \\ V_t \end{pmatrix} = \begin{pmatrix} v_u \\ v_d \end{pmatrix}.$$

As before, since $a \neq b$ the matrix is invertible, so there is a unique solution, which gives the option value at P of $A_t S_t + V_t B_t$ and the hedging strategy. Since the calculation at Q is identical, by uniqueness we obtain the same result. This proves the induction step, and by induction the full claim.

With this observation in mind, we can afford to draw the tree 'recombining': normally that's done by drawing a diagonal grid shape with (up,down) and (down,up) going from left to right round the two sides of a diamond to meet at the right end, rather than drawing a tree where the two paths would lead to two different nodes. We only have to do the above linear algebra computation for the $\sum_{i=0}^{N-1}(i+1)$ nodes in the grid (and to populate the time N nodes, of which there are $N+1$, with the different option values); this is on the order of N^2 calculations. The point is that since the two routes round the diamond get to the same stock price, we only need to compute the node value and hedging strategy once.

Since N^2 is much better than 2^N, this observation makes it somewhat practical to use the binomial model. But only somewhat: if N is a few days and time is measured in milliseconds (the former could well be larger and for some stocks even milliseconds might be a bit unrealistically slow..!) that gives us N on the order of 10^9; while 10^{18} calculations is realistically possible, especially if we use the fact that we can do a lot of calculations in parallel (the different nodes at any given time don't affect each other, so separate processors can be tasked to calculate their values) this is still painfully expensive.

There is a trick we can notice here. We can choose a probability measure to put on our space that gives a useful meaning to our hedging strategy. At the moment, this measure has nothing to do with 'reality', it's simply going to help us with the book-keeping. In particular, we are not assuming that the 'real' stock price movements have anything to do with this measure.

The measure we want to use is the following product measure $\mathbb{Q}$. At each time t, we

assign a probability q to the stock price moving up, and so $1-q$ to it going down, where q satisfies $qb+(1-q)a=r$. Saying that $\mathbb{Q}$ is a product measure is the same thing as saying that these up-or-down events are independent.

The reason this measure is useful is that our wealth is almost a martingale with respect to this measure. We have

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}}\big(A_tS_{t+1}+V_tB_{t+1}\big|\mathcal{F}_t\big) &= A_t\big(q(1+b)S_t+(1-q)(1+a)S_t\big)+V_t(1+r)B_t\\ &= A_tS_t+\big(qb+(1-q)a\big)A_tS_t+(1+r)V_tB_t\\ &= (1+r)\big(A_tS_t+V_tB_t\big)\,.\end{aligned}$$

In order to get an actual martingale, we want to get rid of the annoying $1+r$ factor. We define the *discounted wealth* at time t, by

$$W_t=(1+r)^{-t}\big(A_tS_t+V_tB_t\big)\,.$$

A good way to think about this is that it represents a future wealth at time t in terms of money at time 0. If your wealth at time 0 is y, and you put it all into bonds, then your discounted wealth at time t will simply stay at y.

The above calculation shows that discounted wealth is a martingale with respect to $\mathbb{Q}$. This was the 'point' of our choice of q; the expected change (according to $\mathbb{Q}$, which, again, might have nothing to do with reality..!) in the discounted value between S_t and S_{t+1} is zero.

Theorem 4. *Given $-1<a<r<b<\infty$, and $S_0,B_0,K>0$, and $N\in\mathbb{N}$, the value at time 0 of the European option with maturity N and strike price K is*

$$\mathbb{E}_{\mathbb{Q}}\big((1+r)^{-N}\max(S_N-K,0)\big)\,.$$

Proof. We first prove that discounted wealth $(1+r)^{-n}\big(A_{n-1}S_n+V_{n-1}B_n\big)$, for any choice of $(A_n,V_n)_{n\geq 0}$, is a $\mathbb{Q}$-martingale.

The L_1 property is trivial: for any given n, the σ-algebra $\mathcal{F}_n$ contains only finitely many sets, so that the expected absolute value is a finite sum and hence finite.

The measurability is similarly trivial: all the terms in the above formula are $\mathcal{F}_n$-measurable.

Finally, the martingale property is more or less the calculation we already did, but we repeat it for completeness. Given $n\geq 0$, we have

$$\begin{aligned}&\mathbb{E}_{\mathbb{Q}}\Big((1+r)^{-n-1}\big(A_nS_{n+1}+V_nB_{n+1}\big)\Big|\mathcal{F}_n\Big)\\ &\quad=(1+r)^{-n-1}A_n\big(q(1+b)S_n+(1-q)(1+a)S_n\big)+(1+r)^{-n}V_nB_n\\ &\quad=(1+r)^{-n}\big(A_nS_n+V_nB_n\big)\\ &\quad=(1+r)^{-n}\big(A_{n-1}S_n+V_{n-1}B_n\big)\,,\end{aligned}$$

where the final line is only for $n\geq 1$. The final line (or the penultimate line if $n=0$) is the discounted wealth at time n: the first equality is where we use the definition of $\mathbb{Q}$, the second is by choice of q, and the final line represents the frictionless trading we do immediately after time n if $n\geq 1$; for $n=0$ the penultimate line is our choice of an initial holding of stocks and bonds.

Since we proved that there exists a hedging strategy for the option, taking $(A_n, V_n)_{n\geq 0}$ to be the hedging strategy, we have

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}\Big((1+r)^{-N}\max(S_N-K,0)\Big) &= \mathbb{E}_{\mathbb{Q}}\Big((1+r)^{-N}\big(A_{N-1}S_N+V_{N-1}B_N\big)\Big)\\
&= \mathbb{E}_{\mathbb{Q}}\Big((1+r)^{-N}\big(A_{N-1}S_N+V_{N-1}B_N\big)\Big|\mathcal{F}_0\Big)\\
&= \mathbb{E}_{\mathbb{Q}}\big(A_0S_0+V_0B_0\big)\\
&= x\,,
\end{aligned}$$

where $x = A_0S_0 + V_0B_0$ is the value of the option at time 0. Here the first equality is because (A_n, V_n) is the hedging strategy, the second is because $\mathcal{F}_0$ is the trivial σ-algebra, and the third uses the martingale property of the discounted wealth. □

This calculation suggests we might be able to use our knowledge of probability to find some kind of explicit formula for the value of the option. To do this, we really want to use some kind of calculus, taking some 'limit' as $N \to \infty$, rather than deal with large but finite sums. This route leads to the Black-Scholes theory, and what we need to observe to link things up is that our discounted wealth martingale will be, by Exercise 2, a martingale transform with respect to the q-biased random walk (i.e. Y_i takes value $b-r$ with probability q, and $a-r$ with probability $1-q$); the martingale transform in fact gives us the complete hedging strategy.

However, we can also observe something very nice in this discrete setting. The option price has now been written as an expectation, and with respect to a very easy probability measure to sample from (approximately, at least!). We can use a Monte Carlo estimation to speed up the calculation of the option price. (We didn't cover precisely this estimation in Notes 5; we should really use a concentration inequality which is less sensitive to the largest possible values the integrand can take in favour of computing or bounding the variance; the Bernstein inequalities, for instance, would be appropriate.)

To make this work, observe that we can approximate q by looking at its first s binary digits: if we generate s independent bits (each coming up 1 with probability $\frac{1}{2}$ and zero otherwise) we can think of the s-digit bit string as a number in $[0,1)$; if that number is at least q, then we return Yes, and if not No. If q happens to have only s binary digits (and all the rest are zero) then the probability of Yes is exactly q; if not there is an at most 2^{-s} error. This approach lets us go from independent coin-flips to generating approximately a probability q Bernoulli random variable; we need N of those to do a single Monte Carlo estimate of S_N, which gives us a single Monte Carlo estimate of $\mathbb{E}_{\mathbb{Q}}\big((1+r)^{-N}\max(S_N-K,0)\big)$. We repeat this until we have sufficient confidence in our estimate.

As written, this Monte Carlo estimate is just a way to check what the value of an option is, if you're presented with the option and need to decide whether to purchase it; it doesn't tell you how to construct a hedging strategy. However it's not too hard to modify this approach to obtain a hedging strategy too.

Exercise 7. *Explain why using Monte Carlo estimation to find approximately the discounted value of the option for the two possible stock prices at time* 1 *allows us to decide on the correct hedging strategy at time* 0*. To simplify calculation, assume (which isn't exactly true!) that Monte Carlo estimation does not make any error in approximating the discounted value.*

Explain how to generalise this, using roughly N calculations at each time step, to find the correct hedging strategy for whatever path the stock prices actually take.

Solution to Exercise 7:

This is 'easy' in light of the previous exercise: suppose v_u and v_d are the discounted value of the option at time 1 for the two possible moves, up and down respectively, which we obtain by two Monte Carlo estimations. Then we do the same linear algebra calculation as in the previous exercise to find the (unique) value and hedging strategy for the option at time 0.

Generalising this, the same approach applies for any times $t+1$ and t. Assuming no error is made by the Monte Carlo estimation, the wealth provided by the hedging strategy at time t will precisely match what we need to carry out the required hedging strategy immediately after time t.

This approach doesn't give us a speed up in total compared to the N^2 calculations from the previous exercise: it also needs N^2 calculations in total. However, it does allow us to find a value and a first hedging choice much more quickly, and we therefore gain time to do the next set of N calculations before time 1 arrives.

On the other hand, the assumption 'there are no errors' is unrealistic: you should really check how bad small errors can be, and if they are likely to build up, before implementing this!

To this point, what we just did was purely to use probability for book-keeping and to find a quick way to estimate the (deterministic) option value. Again, we haven't made any assumption in the above that the 'real world' behaviour of S_t has anything to do with the measure $\mathbb{Q}$ we wrote down.

However, if we make an extra assumption, we can actually find an economic interpretation of $\mathbb{Q}$. The assumption is 'efficient markets': we assume that 'the market' knows all relevant information. We also assume that the market prices are 'risk-neutral': (most) players in the market put relatively tiny fractions of their total wealth into any given stock, so that they are indifferent between having a risky stock with a given expected value and simply having that expected value amount of risk-free bonds.

Under this assumption (but still working on the only-up-or-down assumption!), the only way our model can make sense is if 'the market' believes that at any given time, the probability of an up-move really is q: that is, 'the market' believes that the stock price really evolves according to $\mathbb{Q}$. This is the reason that $\mathbb{Q}$ is called the *risk-neutral measure*. The point here is that if you as a player in the market believe something else—maybe you believe that, for a given time and stock price S_t the probability of an up-move coming next is larger than q—then you believe the expected value of S_{t+1} is greater than $(1+r)S_t$. In this case, it makes sense for you to borrow money (go short on bonds), use it to buy stock, and then at the next time period you expect to be able to sell your stock again, repay the borrowed money with interest r, and have profited. Under the efficient markets assumption, if you have a rational cause for your belief, 'the market' should already have acted to price in the cause; the stock price would already have gone up. A similar comment applies if you believe the probability of an up-move is smaller than q.

Finally, suppose that we are working in the context of a wider stock market (but we

currently only care about modelling the fluctuations of our single stock price). We will need to fix on a certain length of time as our 'one time step' for the binomial model. This is a somewhat arbitrary decision; the shorter we set a time step, the more accurate we hope our model will be, but the more computation we need. (This trade-off is something we don't need to worry about if we move to a continuous time model!)

Having made the decision, though, we need to decide what numbers 'make sense' for a and b. This obviously matters: those numbers appear in our formula for the option price. What we can notice is that if a and b are both very close to r, then our stock price S_t is guaranteed stay very close to $(1+r)^t$; in other words, the discounted stock price stays very close to 1. Whereas if a and b are far apart, our discounted stock price will likely move rapidly away from 1. In either case, it follows a geometric random walk without drift, but the variance at any given time depends on whether a and b are close to r, or far apart.

So the choice of a and b is in fact modelling the volatility of our stock. In the continuous limit, there will simply be a single parameter, σ, which is our model for the volatility of the given stock (the justification for this is the Central Limit Theorem: we have two parameters, but we choose q such that the mean is zero: the Central Limit Theorem then tells us that the limit distribution at a given time will only have one parameter, from the variance), and this model parameter volatility affects the modelled option price. In a bigger market model where we model several stocks, with a common time (but we assume their prices are independent!) we might have several different volatility parameters, or several different choices for a and b. Conversely, if we observe a market price for a particular option, we can calculate what 'the market' believes the stock volatility to be (the implied volatility).

What we've done in these notes is a 'deterministic' route to the binomial model option price formula; we set the model up and proved the existence of a hedging strategy before bringing in any probability.

We can make a final observation in this direction: we established that the discounted wealth W_n in our hedging strategy is a martingale adapted to $(\mathcal{F}_n)_{n\geq 0}$, the natural filtration of our process $(Y_i)_{n\in\mathbb{N}}$. By Exercise 2 it follows that the discounted wealth can be written as a martingale transform; but let's do so by writing explicitly the martingale transform. To begin with, we need to find the right reference martingale: $\sum_{i=1}^n Y_i$ is *not* a martingale by summation, because its mean is not zero (unless we happen to have $q=\frac{1}{2}$). The easiest martingale to deal with here is what we get by subtracting the drift (though basically any choice will work; this just gets a relatively clean formula). Let $Z_i = Y_i - (2q-1)$ for each i. Then $\sum_{i=1}^n Z_i$ is a martingale by summation, with the same natural filtration $(\mathcal{F}_n)_{n\geq 0}$, and we want to find the right martingale transform $(C_n)_{n\geq 0}$ which is $(\mathcal{F}_n)$-adapted. It's easiest to start with the discounted stock price change. We have

$$S_n - (1+r)S_{n-1} = \begin{cases} (b-r)S_{n-1} & \text{if } Z_n = 1-(2q-1) \\ (a-r)S_{n-1} & \text{if } Z_n = -1-(2q-1) \end{cases} .$$

Plugging in $2q-1 = \frac{2r-a-b}{b-a}$, from our definition of q, we see $1-(2q-1) = \frac{2b-2r}{b-a}$ and $-1-(2q-1) = \frac{2a-2r}{b-a}$, so this simplifies to

$$S_n - (1+r)S_{n-1} = \tfrac{b-a}{2} S_{n-1} Z_n \,.$$

We can now figure out how the discounted wealth behaves. We have

$$\begin{aligned} W_n - W_{n-1} &= \big(A_{n-1}S_n + V_{n-1}B_n\big)(1+r)^{-n} - \big(A_{n-1}S_{n-1} - V_{n-1}B_{n-1}\big)(1+r)^{1-n} \\ &= A_{n-1}\big(S_n - (1+r)S_{n-1}\big)(1+r)^{-n} \\ &= A_{n-1}\tfrac{b-a}{2}(1+r)^{-n}S_{n-1}Z_n\,, \end{aligned}$$

and this says that our martingale transform is given by $C_{n-1} = A_{n-1}\frac{b-a}{2}(1+r)^{-n}S_{n-1}$ for each $n \in \mathbb{N}$. To rephrase it, the martingale transform C_{n-1} of the discounted wealth captures our hedging strategy—it tells us what holding of stock we need at time $n-1$.

There is another completely different route to these results. Forget that we proved the existence of a hedging strategy. We define the measure $\mathbb{Q}$, and the process $(Z_n)_{n\geq 0}$, as above, and we look at the random variable $W'_N = (1+r)^{-N}\max(0, S_N - K)$. This has a perfectly good interpretation as the discounted value of the the option at time N; in other words, it's trying to write the option value at time N in terms of 'the value of money' at time 0.

Letting (just as above) $(\mathcal{F}_n)_{n\geq 0}$ be the natural filtration of $(Z_n)_{n\geq 0}$, we observe $\sum_{i=1}^n Z_i$ is a martingale by summation, adapted to the filtration.

We can apply the Doob martingale construction: let $W'_n = \mathbb{E}(W'_N|\mathcal{F}_n)$ for each $0 \leq n \leq N$, so in particular $W'_0 = \mathbb{E}\big((1+r)^{-N}\max(0, S_N - K)\big)$. This is by definition a martingale adapted to the filtration. The point is that it turns out to be the same martingale we just discovered as the discounted wealth of the hedging strategy. To see this, we can reverse the calculations we just did. We start by applying Exercise 2 to write W'_n as a martingale transform: there is a unique process $(C'_n)_{n\geq 0}$ which is $(\mathcal{F}_n)$-adapted such that

$$W'_n = W'_0 + \sum_{i=1}^n C'_{i-1}Z_i$$

holds for each $n \geq 0$. We claim that from this we get a hedging strategy with initial value W'_0: let

$$A'_n = C'_n\tfrac{2}{b-a}(1+r)^n S_n^{-1}\,,$$

and we choose V_n to satisfy the conservation of wealth under trading (one can write a formula, but it's not particularly enlightening).

Why is this a hedging strategy? Well, it has the correct value at time N (by definition!), and reversing the calculation we did above shows that it satisfies the definition. In fact, what that calculation shows is that a hedging strategy for the option is basically the same thing as a $\mathbb{Q}$-martingale whose value at time N is the discounted value of the option. We could equally well have used this route from the beginning to prove the existence of a hedging strategy; here we get the uniqueness of the hedging strategy (and the fair price for the option!) from the fact that the martingale transform is unique (and the fact that once we decide on the random variable W'_N, the only $\mathbb{Q}$-martingale whose time N value is W'_N is the Doob construction, because conditional expectations are a.s. equal). Since we have uniqueness, all the 'new' processes with primes we studied via martingale transforms are in fact the same processes as the ones we got to by backwards induction and linear algebra. And in fact, what makes both approaches work is the same linear algebra: here it's done explicitly in Exercise 5, while in the martingale transform approach, it's what makes Exercise 2 work.

Finally, it's worth noting that we really only used this particular 'European option' in order to stick to a recognisable and standard formula. We could equally well obtain fair prices for any other financial product whose value at time N depends only on S_N in this way, for example a product allowing you to buy a unit of stock at time N at price $\frac{1}{2}S_N$. In fact, we can even deal with path-dependent options such as 'Asian options': while the random variable W_N we defined was written as a simple formula of S_N, W_N is permitted to be any $\mathcal{F}_N$-measurable function, so it is allowed to depend on all the values of $Z_1, \ldots, Z_N$.

In the 'path-dependent' setting, we won't any more have the 'recombining' property we used to argue that we can calculate option values in time on the order N^2, and a deterministic calculation of the binomial tree needs exponential time. But the Monte Carlo approach still works.

18 Recap

This course has covered measure theory, in particular as used in probability, in reasonable detail. Of course there is more: we did not prove the Radon-Nikodym theorem, or Fubini's theorem, for example. But we covered what you would need to start looking at continuous time processes formally, or to go in to measure theory in more detail: the 'hard conceptual' part, which is properly understanding what a σ-algebra is and means, how it relates to the intuitive notion of size, and how we can go from one to the other (the Carathèodory Extension Theorem and the construction of pre-measure) was covered in some detail.

We also saw one good example (at least in the notes; we didn't have time in lectures) of why, even over the reals, we should care about a 'better' theory of integration than Riemann: the Law of the Unconscious Statistician is a standard calculation tool that lets us use 'maths methods' integration results to compute expectations of random variables, and we saw that we could prove a fully general statement without too much pain: proofs that don't have the Lebesgue integral available often get significantly harder and still manage only to prove weaker results.

We've also looked at martingales in discrete time in some detail. Here we (deliberately) looked only at one half of the theory: we focused mainly on 'large deviation' inequalities and how to use them, and looked much less at the 'fine detail' side of the theory (though the martingale representation theorem we just saw is a step in this direction). To go further in the direction we did look at, maybe the next step would be to read up on Freedman's inequality; after that the natural next step is to look at large deviations for polynomials rather than linear functions—here one rapidly gets to open problems.

On the 'fine detail' side, a lot is 'missing': perhaps the next things to read are the Doob maximal inequality and the Martingale Convergence Theorem.

We've also seen several applications of these ideas. Maybe the 'nicest' is the Johnson-Lindenstrauss theorem and the binary version; apart from the fact that this is in practice useful, it also illustrates quite well how we can use a continuous setting to do complicated calculations without too much pain, and how we can recover answers for the discrete setting we really want to understand by some 'comparison'. In that case we were 'lucky' in that we could argue the discrete case is strictly better: in most situations this isn't true, but we can often identify a 'continuous limit' of the discrete setting, and attempt to argue that

the discrete setting cannot be too much worse, or converges to the continuous setting in some kind of large N limit. Here we'd need to use results about convergence of random variables and probably things like Hölder's inequality. If we're doing any computations with series (like Taylor series), which is also a standard tool in this area, we probably need either the Monotone or Dominated Convergence Theorem to carry our arguments through (as we did!)—in short, even though we are basically interested in a discrete problem, we'll need a lot of tools from measure theory and the continuous case to get anywhere. It's worth stressing that while we didn't try to show 'bad examples', these most definitely exist: if you are too sloppy about 'everything reasonable is going to converge nicely' in this area, you'll quickly 'prove' things that are actively false; this is why probabilists tend to stress the measure theory basics so much.

The 'differential equations' or 'fluid limit' material we saw is quite useful in a lot of areas; it's also a beautiful application of martingales. In probability, an area this leads to is queueing theory; in computation, this is a tool you can often use to prove performance for more advanced randomised algorithms. The load balancing material is a rather good example of this: what you should think here is that you have a problem to solve, and there are lots of 'good options' but there is also some kind of 'bottleneck' or 'conflict'; if you always choose the same good options early in your computation, you run in to trouble later. You can hope that some kind of random selection will help: and this is what randomised algorithms do. A first course on these will usually focus on ideas that can be made to work without too much trouble on the probabilistic side; for more serious applications, 'differential equations' is a standard tool to prove performance guarantees.

We visited Monte Carlo algorithms and Markov chains rather briefly—enough to explain what those two concepts are, why they often show up together for computation, and nothing more. We also saw only one type of Monte Carlo algorithm (integral estimation): there are more, though this is maybe the biggest area; if you find yourself wanting to do some kind of estimation, you should definitely read up on the area not just apply the two results we saw. Similarly, we didn't do more with Markov chains than vaguely point at why they might be useful for sampling approximately from a desired distribution. If you find yourself wanting to do that, you should never just write down the first Markov chain that comes to mind: read more about this first! However be somewhat cautious about the literature: quite a bit of it (especially written by non-mathematicians) gives a picture of Markov chain convergence as something 'smooth' and gives heuristic 'tests for convergence'. This can be very far from true, as we saw with the infection model (where it 'looks converged' for a very long time without actually being close to the stationary distribution. A 'good' Markov chain won't exhibit this kind of bad behaviour, and you don't know enough to tell which Markov chains might be good or bad yet.

Finally, we gave a reasonable bit of detail on the discrete financial model: the obvious warning is not to even consider applying this until you know what a correct choice for the parameters a and b might be. What we also completely skipped over is any idea of solving these models analytically (as opposed to giving a computational method to find exactly or to estimate numerically the result): the standard next step here is to do it in a continuous time limit, which brings us in to financial mathematics.